

Lecture 26

在 point estimation 中, 我们将关于 parameter θ 和 estimator δ 的 loss function 和 risk function 分别定义为 $L(\theta, \delta)$ 和 $R(\theta, \delta) = E_\theta(L(\theta, \delta)) = \int L(\theta, \delta) dF(x|\theta)$, estimator δ 应 minimize risk.

但在 Bayesian analysis 中, $\theta \in \Theta$ 为 random variable. 考虑 $\theta = \theta_0$ 时的 $L(\theta_0, \delta)$ 和 $R(\theta_0, \delta)$ 并不足够. 因此我们考虑 average risk.

§1 Bayes estimators 和 average risk

1. Definition: Average risk

对于 loss function $L(\theta, \delta)$ 和 risk function $R(\theta, \delta) = E_\theta(L(\theta, \delta))$, the average risk 被定义为:

$$\begin{aligned} r(\Lambda, \delta) &= E(R(\theta, \delta)) \quad (\text{对 } \theta \text{ 取 integral}) \\ &= \int_\Theta R(\theta, \delta) d\Lambda(\theta|\lambda) \\ &= E[E_\theta(L(\theta, \delta))] \\ &= \int_\Theta \int_{\mathcal{X}} L(\theta, \delta) dF(x|\theta) d\Lambda(\theta|\lambda) \end{aligned}$$

其中 $\Lambda(\theta|\lambda)$ 为 prior distribution

注: average risk 其实是关于 λ 的函数, 但我们在 notation $r(\Lambda, \delta)$ 中省去了 λ .

2. Definition: Bayes estimator

任意 minimize average risk $r(\Lambda, \delta)$ 的 estimator δ 被称为 a Bayesian estimator w.r.t. prior Λ

3. Theorem: 一种求 Bayesian estimator 的方法 (可简化计算)

令 $\Theta \ni \theta \in \Theta$ 有 prior distribution $\Lambda(\theta|\lambda)$, 其中 hyperparameter λ 已知.

① 当 $\theta = \theta_0$ 时, X 的分布为 $F_{\theta_0} = f(x|\theta_0)$.

考虑通过非负的 loss function $L(g(\theta), \delta)$ 来估计 $g(\theta)$.

若 ① 存在 estimator δ_0 with finite risk

② 对 almost all x , $\delta_\Lambda(x)$ 能 minimize

$$E[L(g(\theta), \delta(x)) | X=x] \quad (\text{先对 } \theta \text{ 取 integral})$$

则 $\delta_\Lambda(x)$ 为 Bayesian estimator

证明:

若 estimator δ 有 finite risk, 则

$$r(\Lambda, \delta) = E^\theta E^x[L(g(\theta), \delta(x))] < \infty \quad (\text{上标 } x \text{ 表示对 } x \text{ 取期望})$$

$$\Rightarrow r(\Lambda, \delta) = E^x E^\theta[L(g(\theta), \delta(x))] < \infty$$

$$\Rightarrow E^\theta[L(g(\theta), \delta(x))] < \infty \quad \text{a.e.} \quad (\text{由于 } L(\cdot) \text{ 非负})$$

根据 $\delta_\Lambda(x)$ 的定义:

$$E^\theta[L(g(\theta), \delta_\Lambda(x)) | X=x] \geq E^\theta[L(g(\theta), \delta(x)) | X=x] \quad \text{a.e.}$$

$$\Rightarrow E^x E^\theta[L(g(\theta), \delta_\Lambda(x))] \geq E^x E^\theta[L(g(\theta), \delta(x))]$$

$$\Rightarrow r(\Lambda, \delta) \geq r(\Lambda, \delta_\Lambda)$$

4. Theorem: 某些 loss function 的 Bayesian estimator

若上一个 theorem 的 condition 成立, 则

① 令 $L(g(\theta), \delta) = (\delta - g(\theta))^2$ (quadratic loss)

则 $\delta_\Lambda(X) = E(g(\theta)|X)$ (posterior mean)

② 令 $L(g(\theta), \delta) = w(\theta)(\delta - g(\theta))^2$ (weighted quadratic loss)

则 $\delta_\Lambda(X) = \frac{\int w(\theta)g(\theta) \cdot \pi(\theta|X) d\theta}{\int w(\theta) \cdot \pi(\theta|X) d\theta} = \frac{E[w(\theta)g(\theta)|X]}{E[w(\theta)|X]}$ (posterior weighted mean / posterior weight mean)

③ 令 $L(g(\theta), \delta) = |\delta - g(\theta)|$ (absolute loss)

则 $\delta_\Lambda(X)$ 为 $g(\theta)|X$ 的 distribution 的 median (posterior median)

证明: ① 和 ②

由上一个 theorem, 对于 weighted quadratic loss, 我们仅需研究 $\delta_\Lambda(X)$ who minimizes

$$E^\theta[w(\theta)(\delta_\Lambda(X) - g(\theta))^2] = \delta_\Lambda(X)^2 \cdot E(w(\theta)|X) - 2\delta_\Lambda(X) \cdot E(w(\theta)g(\theta)|X) + E(w(\theta)g(\theta)^2|X)$$

注意到为关于 $\delta_\Lambda(X)$ 的 quadratic function, 因此

$$\delta_\Lambda(X) = \frac{E[w(\theta)g(\theta)|X]}{E[w(\theta)|X]}$$

若 $w(\theta) \equiv 1$, 则

$$\delta_\Lambda(X) = E(g(\theta)|X)$$

证明: ③

思路: 证明在 minimize $E[|\delta_\Lambda(X) - g(\theta)| | X=x]$ 层面, posterior median 主要优于其他 $\forall a \neq \xi$

Meanwhile, for absolute loss, we focus on $\delta_\Lambda(x)$ who minimizes

$$\mathbb{E}(|\delta_\Lambda(x) - g(\theta)| | x). \quad (2.5)$$

Define ξ to be any median of the conditional distribution of $g(\theta)$ given $X=x$, in other words, we have

$$\mathbb{P}(g(\theta) > \xi | x) = 1 - \mathbb{P}(g(\theta) \leq \xi | x) \leq \frac{1}{2}. \quad (\text{median 的定义})$$

Notice that for $\forall a \leq \xi$, (先考虑 $a \leq \xi$ 的情况)

$$\begin{aligned} \frac{1}{2} \mathbb{E}(|g(\theta) - a| | x) &= \frac{1}{2} \left[\mathbb{E}((g(\theta) - a) \cdot \mathbb{1}(g(\theta) > a) | x) \right. \\ &\quad \left. - \mathbb{E}((g(\theta) - a) \cdot \mathbb{1}(g(\theta) \leq a) | x) \right] \\ &= \mathbb{E}((g(\theta) - a) \cdot [\mathbb{1}(g(\theta) > a) - \frac{1}{2}] | x). \end{aligned} \quad (2.6)$$

Since $a \leq \xi$, utilize form (2.6), we have

$$\begin{aligned} \frac{1}{2} \mathbb{E}(|g(\theta) - \xi| | x) &= \mathbb{E}((g(\theta) - a) \cdot [\mathbb{1}(g(\theta) > \xi) - \frac{1}{2}] | x) \\ &\quad + (a - \xi) \cdot [\mathbb{P}(g(\theta) > \xi | x) - \frac{1}{2}] \\ &\leq \mathbb{E}((g(\theta) - a) \cdot [\mathbb{1}(g(\theta) > \xi) - \frac{1}{2}] | x). \end{aligned}$$

therefore,

$$\begin{aligned} &\frac{1}{2} \left[\mathbb{E}(|g(\theta) - a| | x) - \mathbb{E}(|g(\theta) - \xi| | x) \right] \\ &\geq \mathbb{E}((g(\theta) - a) \cdot [\mathbb{1}(g(\theta) > a) - \mathbb{1}(g(\theta) > \xi)] | x) \geq 0. \end{aligned}$$

Similarly, for $\forall a \geq \xi$, we have (再考虑 $a \geq \xi$ 的情况)

$$\mathbb{P}(g(\theta) < \xi | x) = 1 - \mathbb{P}(g(\theta) \geq \xi | x) \leq \frac{1}{2},$$

and we have

$$\begin{aligned} \frac{1}{2} \mathbb{E}(|g(\theta) - a| | x) &= \mathbb{E}((g(\theta) - a) \cdot [\frac{1}{2} - \mathbb{1}(g(\theta) < a)] | x), \\ \frac{1}{2} \mathbb{E}(|g(\theta) - \xi| | x) &\leq \mathbb{E}((g(\theta) - a) \cdot [\frac{1}{2} - \mathbb{1}(g(\theta) < \xi)] | x), \end{aligned}$$

which leads to

$$\begin{aligned} & \frac{1}{2} \left[\mathbb{E}(|g(\theta) - a||x) - \mathbb{E}(|g(\theta) - \xi||x) \right] \\ & \geq \mathbb{E} \left((g(\theta) - a) \cdot [\mathbb{1}(g(\theta) < \xi) - \mathbb{1}(g(\theta) < a)] | x \right) \geq 0. \end{aligned}$$

As a summary, we have

$$\mathbb{E}(|g(\theta) - a||x) - \mathbb{E}(|g(\theta) - \xi||x) \geq 0$$

for arbitrary $a \in \mathbb{R}$, meaning that (2.5) is minimized by $\delta_\Lambda(x) = \xi$, i.e., the bayes estimator $\delta_\Lambda(X)$ should be arbitrary median of the conditional distribution of $g(\theta)$ given X . $\square$

例 1: (Poisson - Gamma conjugacy)

• Example 2.5 (**Poisson-Gamma Conjugacy**). The parameter λ of a Poisson distribution is both the mean and the variance of the distribution. Although squared error loss $L_0(\lambda, \delta) = (\lambda - \delta)^2$ is often preferred for the estimation of a mean, some type of scaled squared error loss, for example, $L_k(\lambda, \delta) = (\lambda - \delta)^2 / \lambda^k$, may be more appropriate for the estimation of a variance. If $X_1, \dots, X_n$ are iid Poisson(λ), and λ has the gamma(α, β) prior distribution, then the posterior distribution is

$$\pi(\lambda|x) \propto \lambda^{\sum_{i=1}^n x_i} e^{-n\lambda} \cdot \lambda^{\alpha-1} e^{-\lambda\beta} = \lambda^{(\alpha + \sum_{i=1}^n x_i) - 1} e^{-(\beta + n)\lambda},$$

which is the kernel of Gamma($\alpha + \sum_{i=1}^n x_i, \beta + n$) distribution. Thus, the Bayes estimator under L_k is given by

$$\delta_k(x) = \frac{\mathbb{E}(\lambda^{1-k}|x)}{\mathbb{E}(\lambda^{-k}|x)} = (\beta + n) \left(a + \sum_{i=1}^n x_i - k \right)$$

for $a - k > 0$. Thus, the choice of loss function can have a large effect on the resulting Bayes estimator.

例 2: (Binomial - Beta conjugacy)

令 random sample $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Ber}(p)$, 令 p 的 prior 为 Beta(α, β).

求 quadratic loss 对应的 p 的 Bayesian estimator

(Step 1: 求出 posterior distribution)

$$\begin{aligned} \pi(p|x) & \propto f(x|p) \cdot \pi(p|\alpha, \beta) \\ & \propto \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} \cdot p^{\alpha-1} (1-p)^{\beta-1} \\ & = p^{\alpha + \sum x_i - 1} (1-p)^{\beta + n - \sum x_i - 1} \end{aligned}$$

$$\Rightarrow p|x \sim \text{Beta}(\alpha + \sum x_i, \beta + n - \sum x_i)$$

(Step 2: 求出 posterior mean)

$$\Rightarrow \delta_\Lambda(x) = E[p|x] = \frac{\alpha + \sum x_i}{n + \alpha + \beta} = \underbrace{\frac{\alpha + \beta}{n + \alpha + \beta}}_{\text{mean of prior}} \cdot \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{mean of sample}} + \underbrace{\frac{n}{n + \alpha + \beta}}_{\text{mean of sample}} \cdot \bar{X}$$

注: Bayesian estimator 可以看成 mean of prior 和 mean of sample 的 weighted sum.

① 若 $n \rightarrow \infty$, 则 $\delta_\Lambda(x) \rightarrow \bar{X}$ (此时 $\alpha = \beta = 0$ 为一个 improper prior)

② 若 $\alpha \rightarrow \infty, \beta \rightarrow \infty, \alpha/\beta \rightarrow C < \infty$, 则 $\delta_\Lambda(x) \rightarrow \frac{C}{1+C}$

例 3: (Binary classification)

令 random sample $X_1, \dots, X_n$ 取自 f_0 或 f_1 , 即 $X_i \stackrel{i.i.d.}{\sim} f_{\theta_i}, \theta_i \in \Theta = \{0, 1\}$,

令每个 θ_i 的 prior 为 Ber(p).

求 0-1 loss $L(\theta_i, \delta_i) = 1\{\delta_i \neq \theta_i\}$ 对应的 θ_i 的 Bayesian estimator

(Step 1: 求出 posterior distribution)

$$\begin{aligned}\pi(\theta_i | x) &\propto f_{\theta_i}(x_i) \cdot p^{\theta_i} (1-p)^{1-\theta_i} \\ &= f_0(x_i) \cdot (1-p) \cdot 1\{\theta_i=0\} + f_1(x_i) \cdot p \cdot 1\{\theta_i=1\}\end{aligned}$$

(Step 2: 写出 risk, 求出 minima)

$$r(\theta_i, \delta_i) = E^{\theta_i} E^{x_i} [1\{\delta_i \neq \theta_i\}]$$

$\Rightarrow$ 根据 theorem, 仅需 minimize $E[1\{\delta_i \neq \theta_i\} | x]$

$$E[1\{\delta_i \neq \theta_i\} | x] = f_0(x_i) \cdot (1-p) \cdot 1\{\delta_i \neq 0\} + f_1(x_i) \cdot p \cdot 1\{\delta_i \neq 1\}$$

$\Rightarrow \begin{cases} \text{若 } f_0(x_i) \cdot (1-p) \geq f_1(x_i) \cdot p, \text{ 选取 } \delta_i = 0 \\ \text{若 } f_0(x_i) \cdot (1-p) < f_1(x_i) \cdot p, \text{ 选取 } \delta_i = 1 \end{cases}$ (δ_i 也可取其他值, 但为了有意义, 令 $\delta_i \in \{0, 1\}$)

$$\Rightarrow \delta_i = 1 \left\{ \frac{f_1(x_i) \cdot p}{f_0(x_i) \cdot (1-p)} \geq 1 \right\}$$

例 4: (Normal mean Bayes estimator)

令 random sample $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} N(\theta, \sigma^2)$, 其中 σ 已知. 令 θ 的 prior 为 $N(\mu, \tau^2)$.
求 quadratic loss 对应的 θ 的 Bayesian estimator

(Step 1: 求出 posterior distribution)

$$\begin{aligned}\pi(\theta | x) &\propto f(x | \theta) \cdot \Lambda(\theta | \mu, \tau^2) \\ &\propto \exp\left(-\frac{1}{2\sigma^2} \sum (x_i - \theta)^2\right) \cdot \exp\left(-\frac{1}{2\tau^2} (\theta - \mu)^2\right) \\ &\propto \exp\left(-\frac{1}{2\sigma^2} n\theta^2 + \frac{n\bar{x}}{\sigma^2} \theta\right) \cdot \exp\left(-\frac{1}{2\tau^2} \theta^2 + \frac{\mu}{\tau^2} \theta\right) \\ &\propto \exp\left\{-\frac{1}{2} \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right) \left(\theta - \frac{n\bar{x}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2}\right)^2\right\}\end{aligned}$$

$$\Rightarrow \theta | x \sim N\left(\frac{n\bar{x}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2}, \frac{1}{n/\sigma^2 + 1/\tau^2}\right)$$

(Step 2: 求出 posterior mean)

$$\Rightarrow \delta_{\Lambda}(x) = E[\theta | x] = \frac{n\bar{x}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2} = \underbrace{\frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2} \cdot \bar{X}}_{\text{mean of sample}} + \underbrace{\frac{1/\tau^2}{n/\sigma^2 + 1/\tau^2} \cdot \mu}_{\text{mean of prior}}$$

3. Generalized Bayes Estimator and the Limit of Bayes Estimator

Let δ be an estimator. We say that δ is a generalized Bayes estimator if there exists a probability measure ν on Θ such that δ is a Bayes estimator with respect to the posterior distribution $\pi(\theta|x)$ obtained using the improper prior ν . Such a posterior is also called a formal posterior.

Definition 3.1 (♣ Generalized Bayes Estimator). An estimator $\delta_\Lambda(X)$ is called a generalized Bayes estimator if it is a Bayes estimator with respect to a posterior distribution obtained using an improper prior, and such a posterior is also called a formal posterior.

Definition 3.2 (♣ The Limit of Bayes Estimator). An estimator $\delta(X)$ is a limit of Bayes estimator if there exists a sequence of proper priors $\{\Lambda_k\}_{k \geq 1}$

and bayes estimator $\{\delta_{\Lambda_k}\}_{k \geq 1}$ such that when $k \rightarrow \infty$,

$$\delta_{\Lambda_k}(X) \xrightarrow{a.s.} \delta(X), \quad \text{i.e., } \delta_{\Lambda_k}(x) \rightarrow \delta(x), \quad a.e..$$

• **Example 3.3 (♣ Pitman Estimator: the Generalized Bayes Estimator of Location Family)**. Let $X_1, \dots, X_n$ be a random sample from a location family $f(x - \theta)$ with a known function f . Assign $\Lambda(\theta) \equiv 1$ as an improper prior for θ . Give the bayes estimator of θ with respect to the quadratic loss.

$$\begin{aligned} X_1, \dots, X_n &\stackrel{i.i.d.}{\sim} f(x - \theta), \quad \Lambda(\theta) \equiv 1 \\ \Rightarrow \pi(\theta|X) &\propto \prod_{i=1}^n f(x_i - \theta) \\ \Rightarrow \delta_{\Lambda}(X) = E[\theta|X] &= \int \theta \cdot \pi(\theta|X) d\theta = \frac{\int \theta \cdot \prod_{i=1}^n f(x_i - \theta) d\theta}{\int \prod_{i=1}^n f(x_i - \theta) d\theta} \end{aligned}$$

• **Example 3.4 (♣ Limit of Bayes Estimator in Binomial-Beta Conjugacy)**. Recall that in the Binomial-Beta Conjugacy, the bayes estimator corresponding to the $\text{Beta}(\alpha, \beta)$ is

$$\delta_{\Lambda}(X) = \frac{\alpha + \sum_{i=1}^n X_i}{n + \alpha + \beta},$$

Therefore, if we consider a sequence of prior $\{\Lambda_k\}_{k \geq 1}$,

$$\Lambda_k = \text{Beta}(\alpha_k, \beta_k), \quad \text{where } \alpha_k = \frac{1}{k}, \quad \beta_k = \frac{1}{k}.$$

Therefore, if we define δ_{Λ_k} as the bayes estimator corresponding to the prior Λ_k , consider the limit of bayes estimator δ , which is given by

$$\delta = \lim_{k \rightarrow \infty} \delta_{\Lambda_k} = \lim_{k \rightarrow \infty} \frac{\frac{1}{k} + \sum_{i=1}^n X_i}{\underbrace{n + \frac{2}{k}}_{\rightarrow n}} = \bar{X}.$$

Hence $\bar{X}$ is a limit of bayes estimator. *考虑 limit of Bayesian statistic 时, n 可被视作 constant*

Recall that an estimator δ is admissible if it is not uniformly dominated by some other estimator, and δ is inadmissible if and only if there exists δ' , such that $R(\theta, \delta') \leq R(\theta, \delta)$ for all $\theta \in \Theta$ and $R(\theta, \delta') < R(\theta, \delta)$ for some $\theta \in \Theta$. Surprisingly, the most existing property of the Bayes estimator is it ensures admissibility. More specifically,

Theorem 3.5 (♣ Admissibility of the Bayes Estimator). *If a bayes estimator is unique, then it is admissible.*

Proof. Denote δ_Λ to be the bayes estimator corresponding to prior Λ . If there exists an estimator δ' such that $R(\theta, \delta') \leq R(\theta, \delta_\Lambda)$ for all $\theta \in \Theta$. Then,

$$r(\Lambda, \delta') = \int_{\Theta} R(\theta, \delta') d\Lambda(\theta) \leq \int_{\Theta} R(\theta, \delta_\Lambda) d\Lambda(\theta) = r(\Lambda, \delta_\Lambda)$$

But $\delta_\Lambda = \arg \min_{\delta} r(\Lambda, \delta)$ according to the definition of bayes estimator. Thus we have $r(\Lambda, \delta') = r(\Lambda, \delta_\Lambda)$, and since we require bayes estimator to be unique, therefore, we have $\delta' = \delta_\Lambda$ with probability 1, meaning that this bayes estimator δ_Λ is admissible. $\square$

This theorem naturally raises the question of when will a bayes estimator δ_Λ be unique. As a result,

* **Theorem 3.6 (♣ Uniqueness of the Bayes Estimator).** Denote Q to be the marginal distribution of X , i.e.,

$$Q(x) = \int f(x|\theta) d\Lambda(\theta).$$

if the average risk $r(\Lambda, \delta)$ defined with respect to a strictly convex loss function is finite for some δ , and $f(\cdot|\theta)$ is absolutely continuous with respect to Q for all θ . Then the bayes estimator δ_Λ who minimize the average risk is unique.

A proof of Theorem.3.6 is provided in [Lehmann and Casella \(2006\)](#).

Further, under certain conditions, the converse is also true.

Theorem 3.7 (♣ Bayesian Characterization of the Admissible Estimator). Let $X \sim f(x|\theta)$ be a density with support $\mathbb{R}$ for every $\theta \in \Theta$. Let the loss function $L(\theta, \delta)$ be continuous, strictly convex in δ for every θ , and satisfy

$$\lim_{|\delta(x)| \rightarrow \infty} L(\theta, \delta(x)) = \infty \quad \text{for all } \theta \in \Theta.$$

Then, to every admissible estimator $\delta(X)$ there exists a sequence $\{\Lambda_k\}_{k \geq 1}$ of prior distributions with support on a finite set (and hence with finite average risk) for which

$$\delta_{\Lambda_k}(X) \xrightarrow{a.s.} \delta(X),$$

where $\delta_{\Lambda_k}(X)$ is the bayes estimator corresponding to prior Λ_k .

A proof of Theorem.3.7 is provided in [Brown \(1986\)](#).

Theorem 3.8 (♣ Bayesian Characterization of the Admissible Estimator in Exponential Family). Under the condition of Theorem.3.7, if we have $\{f(x|\theta), \theta \in \Theta\}$ forms an exponential family, then any admissible estimator is a generalized bayes estimator. In other words, $\delta(X)$ and Λ_k in theorem.3.7 satisfy

$$\Lambda_k(\theta) \rightarrow \Lambda, \quad \text{a.e.,}$$

for some improper prior Λ and $\delta(X) = \delta_\Lambda(X)$ is the generalized bayes estimator for this improper prior Λ .

A proof of Theorem.3.8 is provided in Berger and Srinivasan (1978).

• **Example 3.9 (♣ Admissibility of $\bar{X}$ in Normal Distribution)**. Give the admissibility of $\bar{X}$ for estimating the mean of a normal distribution $N(\mu, \sigma^2)$ with a known variance σ^2 .

Answer. Consider a sequence of prior $\{\Lambda_k(\mu)\}_{k \geq 1}$, we assign $\Lambda_k(\mu) \sim N(\eta, \tau_k^2)$ with a known η and τ_k^2 , where we require $\tau_k \rightarrow \infty$. Then according to Λ_k , the marginal distribution of X is

$$\begin{aligned} m(x) &= \int \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(- \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2} \right) \cdot \frac{1}{\sqrt{2\pi\tau_k^2}} \exp \left(- \frac{(\mu - \eta)^2}{2\tau_k^2} \right) d\mu \\ &\propto \exp \left(- \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{2\sigma^2} \right) \int \exp \left(- \frac{(\bar{x} - \mu)^2}{2\sigma^2} - \frac{(\mu - \eta)^2}{2\tau_k^2} \right) d\mu \\ &\propto \exp \left(- \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{2\sigma^2} \right) > 0. \end{aligned}$$

and since $f(x|\mu) > 0$, so both $m(\cdot)$ and $f(\cdot|\mu)$ has support $\mathbb{R}$ and implies that $f(\cdot|\mu)$ is absolutely continuous with respect to $m(\cdot)$ for all μ . Thus the bayes estimator of estimating μ is unique, and according to example.??, the bayes estimator $\delta_{\Lambda_k}(X)$ is given by

$$\delta_{\Lambda_k}(X) = \frac{n\bar{X}/\sigma^2 + \eta/\tau_k^2}{n/\sigma^2 + 1/\tau_k^2} = \left(\frac{n/\sigma^2}{n/\sigma^2 + 1/\tau_k^2} \right) \bar{X} + \left(\frac{1/\tau_k^2}{n/\sigma^2 + 1/\tau_k^2} \right) \eta,$$

so by theorem.3.5, $\delta_{\Lambda_k}(X)$ is admissible for estimating μ . Notice that, for a fixed sample size n , when we ask $k \rightarrow \infty$, we have $\delta_{\Lambda_k}(X) \xrightarrow{a.s.} \delta_0(X) = \bar{X}$.

proof by def.

Now, if $\delta_0(X)$ is not admissible for estimating μ , then there exists an estimator $\delta^*(X)$ for which may depend on sample size n but not dependent on k , such that

$$R(\mu, \delta^*) \leq R(\mu, \delta_0) \text{ for } \forall \mu \in \mathbb{R}, \quad \text{and} \quad R(\mu_0, \delta^*) < R(\mu_0, \delta_0) \text{ for some } \mu_0 \in \mathbb{R}.$$

Since $(R(\mu, \delta^*) - R(\mu, \delta_0))$ is a continuous function μ , therefore, there exists some $\epsilon = (R(\mu_0, \delta_0) - R(\mu_0, \delta^*))/2 > 0$, and some corresponding μ_1 and μ_2 , such that

$$R(\mu, \delta_0) - R(\mu, \delta^*) > \epsilon \quad \text{for all } \mu_1 \leq \mu \leq \mu_2.$$

Thus, corresponding to $\Lambda_k(\mu)$, for the average risk function, we have

$$\begin{aligned} \frac{r(\Lambda_k, \delta_0) - r(\Lambda_k, \delta^*)}{r(\Lambda_k, \delta_0) - r(\Lambda_k, \delta_{\Lambda_k})} &= \frac{\int (R(\mu, \delta_0) - R(\mu, \delta^*)) \Lambda_k(\mu) d\mu}{\mathbb{E}[\mathbb{E}[(\bar{X} - \mu)^2 | \mu]] - \mathbb{E}[\mathbb{E}[(\delta_{\Lambda_k} - \mu)^2 | \mu]]} \\ &\geq \frac{\int_{\mu_1}^{\mu_2} \epsilon \cdot \Lambda_k(\mu) d\mu}{\frac{\sigma^2}{n} - \frac{1}{n/\sigma^2 + 1/\tau_k^2}} = \frac{n(n\tau_k + \sigma^2/\tau_k)\epsilon}{\sigma^4 \sqrt{2\pi}} \int_{\mu_1}^{\mu_2} \exp \left(- \frac{(\mu - \eta)^2}{2\tau_k^2} \right) d\mu \end{aligned}$$

when we fix the sample size n and ask $k \rightarrow \infty$, i.e., ask $\tau_k \rightarrow \infty$, we conclude that the integrand converges to $(\mu_2 - \mu_1)$ and

$$\frac{r(\Lambda_k, \delta_0) - r(\Lambda_k, \delta^*)}{r(\Lambda_k, \delta_0) - r(\Lambda_k, \delta_{\Lambda_k})} \rightarrow \infty.$$

which means there exists some k , such that $r(\Lambda_k, \delta^*) < r(\Lambda_k, \delta_{\Lambda_k})$. But this contradicts with δ_{Λ_k} being the bayes estimator with respect to Λ_k . Hence our assumption of the existence of such δ^* is wrong and $\bar{X}$ is admissible for estimating μ . $\square$

This type of approach for proving the admissibility is called the **Blyth's Method**. The procedure can be summarized as follows. **First, we seek for a sequence of prior distribution $\{\Lambda_k\}_{k \geq 1}$** and hopefully, the corresponding bayes estimator δ_{Λ_k} would converge to the estimator δ_0 for which we want to establish its admissibility. **Second, we investigate the rate of $r(\theta, \delta_0) - r(\theta, \delta_{\Lambda_k})$ converge to zero.** **Third, we proof by contradiction and assume there exists an estimator δ^* can dominate δ_0 ,** and utilize the continuity of the risk function to conclude $R(\theta, \delta^*) < (R(\theta, \delta_0) - \epsilon)$ for a positive measure set A . **Fourth, we investigate the rate of $r(\theta, \delta_0) - r(\theta, \delta^*)$,** if it converge to zero in a smaller rate, then we would conclude $r(\theta, \delta^*) < r(\theta, \delta_{\Lambda_k})$ and get a contradiction, which further proofs the admissibility of δ_0 .

- *Example 3.10* (**♣ Admissibility in one-parameter Gamma Distribution**). For a random sample $X = \{X_1, \dots, X_n\}$ from a Gamma distribution $\Gamma(\alpha, \beta)$ with a known α . Show the admissibility of $(\sum_{i=1}^n X_i)/(n\alpha + 1)$ for estimating $\gamma = 1/\beta$ with the scale invariant loss function $L(\gamma, \delta) = (\gamma - \delta)^2/\gamma^2$.

4. Hierarchical Bayes

For a large and complicated model, more than often, despite the fact that our interests lies in the parameter θ , the hyperparameters γ appear to have some random variable's feature and the data are generated from a multi-layer model. Consequently, rather than specifying the prior distribution as a single function, we specify it in a hierarchy, and commonly, a hierarchical bayes model can be formed as

$$X|\theta \sim f(x|\theta), \quad \theta|\gamma \sim \Lambda(\theta|\gamma), \quad \gamma \sim \tilde{\Lambda}(\gamma). \quad (4.1)$$

Regarding (4.1), the posterior is

$$\begin{aligned}\pi(\theta|x) &= \int \pi(\theta, \gamma|x) d\gamma = \int \frac{\pi(x, \theta, \gamma)}{m(x)} d\gamma = \int \pi(\theta|\mathbf{x}, \gamma) \pi(\gamma|\mathbf{x}) d\gamma \\ &= \left[\int f(x|\theta) \cdot \Lambda(\theta|\gamma) \cdot \tilde{\Lambda}(\gamma) d\gamma \right] / \left[\int \int f(x|\theta) \cdot \Lambda(\theta|\gamma) \cdot \tilde{\Lambda}(\gamma) d\gamma d\theta \right].\end{aligned}$$

Therefore, for a given loss function $L(\theta, \delta)$, the average risk is

$$r(\Lambda, \tilde{\Lambda}, \delta) = \int L(\theta, \delta) \pi(\theta|x) d\theta = \int \left[\int L(\theta, \delta) \pi(\theta|x, \gamma) d\theta \right] \pi(\gamma|x) d\gamma.$$

If the loss function is given by quadratic loss $L(\theta, \delta) = (\delta - \theta)^2$, then theorem.2.4 states that the bayes estimator minimize the average risk conditional on γ , i.e.,

$$\arg \min_{\delta} \int L(\theta, \delta) \pi(\theta|X, \gamma) d\theta = \mathbb{E}(\theta|X, \gamma),$$

while the bayes estimator minimize the average risk $r(\Lambda, \tilde{\Lambda}, \delta)$ is

$$\delta_{\Lambda, \tilde{\Lambda}} = \mathbb{E}(\theta|X) = \mathbb{E}[\mathbb{E}(\theta|X, \gamma)|X],$$

which shows that the hierarchical bayes estimator can be thought of as a mixture of single-prior bayes estimator.

• *Example 4.1* (**♣ Conjugate Normal Hierarchy**). Consider a hierarchical bayes model with normal conjugate prior

$$\begin{aligned}X_i|\theta &\sim N(\theta, \sigma^2), \quad \text{with a known } \sigma^2 \text{ and } i = 1, \dots, n, \\ \theta|\tau &\sim N(0, \tau^2), \quad \frac{1}{\tau^2} \sim \text{Gamma}\left(\frac{\nu}{2}, \frac{\nu}{2}\right), \quad \text{with a known } \nu.\end{aligned}$$

Give the bayes estimator of θ .